

Velocity Potential

For an irrotational flow (**Vorticity is zero**)

$$\xi = \nabla \times \vec{V} = 0$$

Consider the following vector identity: if ϕ is a scalar function, then

$$\nabla \times (\nabla \phi) = 0$$

Comparing the two relations, we get

$$\vec{V} = \nabla \phi$$

For an irrotational flow, there exists a scalar function ϕ such that the velocity is given by the gradient of ϕ .

From the definition of the gradient in cartesian coordinates,

$$u \hat{i} + v \hat{j} + w \hat{k} = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

Thus, in cartesian coordinates,

$$u = \frac{\partial \phi}{\partial x}; v = \frac{\partial \phi}{\partial y}; w = \frac{\partial \phi}{\partial z}$$

The velocity potential is analogous to the stream function in the sense that both can be used to find the velocities. However, there are distinct differences between velocity potential and stream function.

1. The flow velocities are obtained by differentiating ϕ in the same direction as the velocities, whereas, ψ is differentiated normal to the velocity direction.

Velocity potential (ϕ):

$$u = \frac{\partial \phi}{\partial x}; v = \frac{\partial \phi}{\partial y}; w = \frac{\partial \phi}{\partial z}$$

Stream function (ψ):

$$\rho u = \frac{\partial \psi}{\partial y}; \rho v = -\frac{\partial \psi}{\partial x} \text{ or } u = \frac{\partial \psi}{\partial y}; v = -\frac{\partial \psi}{\partial x}$$

2. The **velocity potential** is defined for **irrotational flow** only, whereas, the stream function can be used in either rotational or irrotational flows.
3. The **velocity potential** applies to **three-dimensional flows**, whereas the **stream function** is defined for **two-dimensional flows** only.

Special case: **Stream function** can be defined for **axisymmetric flows**, such as the flow over a cone at zero degrees angle of attack. However, for such flows, only **two spatial coordinates** are needed to describe the flow field.

Reference:

Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.